

Anvaya of Mathematical Definitions

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Chapter 1

Set Theory and Real Numbers

1.1 Basic Set Theory

We begin our journey into mathematical analysis with fundamental concepts about sets.

Definition 1.1.1

A **set** is a collection of distinct objects called *elements*. If x is an element of set A , we write $x \in A$.

Explanation

For example, $A = \{1, 2, 3\}$ is a set containing three elements. We have $2 \in A$ but $5 \notin A$.

1.1.1 Countable and Uncountable Sets

One of the most important distinctions in mathematics is between countable and uncountable sets.

Definition 1.1.2: Countable Set

A set S is **countable** if there exists a bijection between S and a subset of the natural numbers $\mathbb{N}$.

Explanation

This means we can "list" all elements of a countable set, even if the set is infinite. The natural numbers $\mathbb{N}$ and integers $\mathbb{Z}$ are both countable.

Theorem 1.1.1: Integers are Countable

The set of integers $\mathbb{Z}$ is countable.

Explanation

We can prove this by showing the bijection $f : \mathbb{N} \rightarrow \mathbb{Z}$ defined by:

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ -\frac{n+1}{2} & \text{if } n \text{ is odd} \end{cases}$$

This gives us the sequence: $0, -1, 1, -2, 2, -3, 3, \dots$

1.2 The Real Numbers

Definition 1.2.1: Complete Ordered Field

The real numbers $\mathbb{R}$ form a complete ordered field, satisfying the completeness axiom: every non-empty set that is bounded above has a least upper bound.

Theorem 1.2.1: Uncountability of Reals

The set of real numbers $\mathbb{R}$ is uncountable.

Explanation

This fundamental result shows there are "more" real numbers than natural numbers. Cantor's diagonal argument provides an elegant proof of this fact.

Chapter 2

Sequences and Convergence

2.1 Convergent Sequences

Definition 2.1.1: Sequence Convergence

A sequence (a_n) **converges** to limit L if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have $|a_n - L| < \varepsilon$.

Lemma 2.1.1: Squeeze Theorem

If $a_n \leq b_n \leq c_n$ for all n , and $\lim a_n = \lim c_n = L$, then $\lim b_n = L$.

Explanation

The Squeeze Theorem is particularly useful for finding limits of sequences that are "trapped" between two other sequences with known limits.

Chapter 3

Continuous Functions

3.1 Definition of Continuity

Definition 3.1.1: Continuity

A function $f : D \rightarrow \mathbb{R}$ is **continuous at point** c if for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $|x - c| < \delta$, we have $|f(x) - f(c)| < \varepsilon$.

Explanation

Intuitively, continuity means we can draw the graph of the function without lifting our pen. Small changes in input produce small changes in output.

As we saw in Theorem 1.2.1, the real numbers are uncountable, which has profound implications for continuous functions on $\mathbb{R}$.